

The Benefits of Random Variable Practice for Spatial Accuracy and Error Detection in a Rapid Aiming Task

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The role of blocked and random variable practice in the development of a spatial error detection mechanism in 2 experiments was investigated in the present study. Twenty-four (Experiment 1) and 40 (Experiment 2) college-aged participants made either 20, 40, or 60° quick lever reversal movements in the sagittal plane. During acquisition in both experiments, blocked practice resulted in less spatial absolute constant error (|CE|) relative to the random group. The blocked practice group showed a smaller mean absolute objective-subjective difference than the random practice group during acquisition (Experiment 1). On no-knowledge of results retention tests in both studies, the random practice group showed a smaller spatial |CE| and a smaller mean absolute objective-subjective difference than the blocked practice group, even on a novel amplitude (Experiment 2). The study demonstrated the advantage of random variable practice over blocked variable practice on retention tests for spatial performance and in developing a spatial error detection capability.

Key words: accuracy, contextual interference, practice organization, recognition capability

The capability of individuals to detect and correct movement errors based on response-produced feedback has long been recognized as a crucial element in the learning of motor skills. Learners who could evaluate intrinsic feedback and correct their own errors could continue to improve without the constant need of a teacher or a coach. In fact, the capability to successfully detect and correct movement errors was a milestone which signaled the passage from a novice to a more advanced stage of motor learning (Adams, 1971).

Naturally, the error detection and correction process became a central issue in the early theories of motor learning (Adams, 1971; Schmidt, 1975). In Adams' (1971) closed-loop theory, participants concurrently evaluated response-produced feedback during slow movements and halted the limb when intrinsic feedback matched the perceptual trace, a feedback-based internal referent of the target position. In Schmidt's (1975) schema theory, participants paired response-produced feedback with knowledge of the movement outcome to form the recognition schema which was responsible for the evaluation of future movements. Learning of the perceptual trace and the recognition schema was maximized, according to both theories, by

initial practice with knowledge of results (KR). Without KR the strength of the perceptual trace would be degraded due to inaccurate motor responses, and the recognition schema would be weakened by the lack of outcome information. Although both theories agreed on the role of KR in the learning process, they disagreed on how practice should be structured. Adams felt that the perceptual trace should be developed with constant, repeated practice of a given target location. However, schema theory (Schmidt, 1975) holds that practicing different outcomes with the same motor program (i.e., variable practice) would result in a stronger schema than constant practice (for more details, see Lee, Magill, & Weeks, 1985; Shapiro & Schmidt, 1982; and Van Rossum, 1990, for reviews of schema theory).

Research that followed the publication of Adams' (1971) or Schmidt's (1975) theory demonstrated clear evidence for the existence of an error detection mechanism. For example, Schmidt and White (1972) demonstrated that objective-subjective correlations (i.e., between the participant's actual response and the subjective evaluation of their own response) increase with practice of a rapid timing movement, suggesting that the error detection mechanism develops with practice. Newell (1974) demonstrated that the mean absolute objective-subjective difference decreased over practice trials of a rapid timing movement, also indicating an enhanced error detection capability. Generally, the greater the number of practice trials, the better the error detection capability (Adams & Goetz, 1973; Adams, Goetz, & Marshall, 1972).

Although there is ample evidence supporting the existence of an error detection mechanism, the ques-

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tion of how the practice session should be organized to maximize the development of the error detection capability of a participant has not been fully addressed. In one of the few studies in the area, Williams and Rodney (1978) showed that practicing random over- and undershoots around a criterion location improved production of the criterion relative to a group that experienced a sequence of decreasing over- and undershoots. However, Stelmach and Szendrovits (1981) showed that error detection and error correction were more effective following sequential, rather than random, practice of five target locations. However, in both the Williams and Rodney and the Stelmach and Szendrovits studies, the use of slow positioning movements prevented a clear distinction between movement production and error detection processes. Finally, Newell and Shapiro (1976) found that participants estimated their movement error for a novel rapid timing task more effectively after a fast-slow blocked practice order compared to the reverse order.

One major limitation of the studies mentioned here is that the type of variable practice, random or blocked, was not manipulated. Research has shown that random practice of different motor skills results in better long-term retention than blocked practice (i.e., Lee & Magill, 1983; Shea & Morgan, 1979), so one might wonder whether practice organization of this type is important for developing error detection capability. Because the effect of practice organization on the development of error detection mechanisms has not been fully investigated, the present investigation intended to extend the literature in this area.

Specifically, the present experiments used the same random-blocked manipulation as in earlier studies but asked whether or not the increased information-processing demands created by random practice apply both to error recognition and movement production processes. During random practice participants must be able to discriminate between the various motor plans to perform a given skill on demand. But if participants attempt to estimate their errors during practice, they must be able to compare response-produced feedback to a sensory template of one of several motor skills or different versions of the same program present in working memory. They might be forced to focus on the sensory differences between the movements to allow for accurate error detection, perhaps leading to better recognition processes. Therefore, the increased information-processing demands of random practice that result in better movement production in long-term retention would also result in better error detection as well.

On the basis of the notions here, one would expect participants to have better error detection capability after random practice compared to blocked practice. However, if the increased information-processing de-

mands of random practice are limited to motor production processes, then there should be no advantage to random practice over blocked practice in developing error detection capability.

Experiment 1

Method

Participants

The participants were 24 undergraduate kinesiology students (16 women and 8 men, mean age = 21.2 years, $SD = 2.3$) at the University of Colorado. Participants were right-handed (based on limb preference for handwriting) and received course credit for their participation. Written informed consent was obtained from the participants prior to inclusion in the study.

Apparatus

The apparatus was a Plexiglas platform on a tabletop, which was slotted to allow one aluminum hand lever (16 cm in length) to move 75° in the sagittal plane, with the most proximal position called 0° . A Beckman potentiometer (Model No. 3351, Beckman Instruments, Brea, CA) was affixed to the base of the lever so displacement could be recorded. The output of the potentiometer was digitized on-line at 250 Hz and stored on hard disks by a MINC 11/23 computer (Digital Equipment Corporation, Merrimack, NH).

Task

The task for the participant was to make a rapid lever reversal movement with the right limb so the reversal point was either 20° , 40° , or 60° . The goal time to reversal was 225 ms. Previous work (Sherwood, 1994) has shown that the relative timing of this task remains constant across changes in amplitude and time to reversal, so participants simply needed to change a force or amplitude parameter to complete the three versions of the task and engage in variable practice (Schmidt, 1988). The starting point for all movements was 0° . The movement required extension and flexion of the elbow joint and a slight forward flexion and extension at the glenohumeral joint. No wrist action was involved. Participants were asked to move to the reversal point and back in a smooth manner without waiting or hesitating at the reversal point. Participants were prevented from viewing their hands by placing a frame-supported opaque sheet over the apparatus and the participants' arms.

Experimental Design

A mixed experimental design was employed with the participants quasi-randomly assigned (to equate the number of men and women in each group) to either random ($n = 12$) or blocked ($n = 12$) practice during acquisition. Participants in the blocked practice group consecutively practiced 30 trials of each amplitude. The order of the amplitudes was counterbalanced in the blocked practice group so that 2 participants practiced each of the six possible orders. Participants in the random practice group practiced a different amplitude on each trial with the restriction that a given amplitude was never repeated on consecutive trials. The 90 trials were analyzed in blocks of 10 trials, forming the repeated measure factor. Participants were not told that they would have no-KR retention tests following the acquisition trials.

In order to assess any effect of practice specificity, participants were also randomly assigned to either a random or blocked practice condition for both no-KR retention tests, thereby forming four independent groups with 6 participants in each group (random/random, random/blocked, blocked/blocked, and blocked/random) and completing a double transfer design. Participants performed 30 trials (10 for each amplitude) 2 min following acquisition in an immediate retention test and 24 hr later in a delayed retention test. Participants were asked to guess their reversal point 5 s following each movement. The order of the amplitudes was counterbalanced in the blocked practice group so that each of the six possible orders was presented to 2 participants during the retention tests. The assignment of random or blocked retention test conditions and the order of conditions for blocked participants were maintained across both retention tests.

Procedure

Each trial began with the participant sitting in front of the apparatus and grasping the upper portion of the lever so that the upper arm was vertical and the elbow joint was 90°. Before each trial the experimenter announced the goal amplitude for the upcoming trial. On a brief auditory stimulus, the participant attempted to move the lever to the goal reversal point and back to the starting position. Five seconds after the completion of the movement, the experimenter asked the participant to guess the reversal point of the just completed movement to the nearest degree. The participant's estimated reversal point (i.e., the subjective score, *S*) was recorded, and the experimenter then gave KR about the actual location of the reversal point in degrees (i.e., the objective score, *O*). Participants also received qualitative KR in time to reversal using a 10% bandwidth such

that they were told "Okay" on times between 202 and 247 ms, "Too fast" on times less than 202 ms, and "Too slow" on times greater than 247 ms. The intertrial interval was approximately 10 s. Quantitative KR was given for spatial error to help the participant more precisely "calibrate" their spatial error detection mechanism and to help the participant focus their attention on the spatial aspects of the task. Qualitative KR was given for time to reversal to reduce the task difficulty and to draw the participant's attention away from the timing aspect of the task which was secondary in importance to the spatial aspects.

Data Analysis

Spatial performance was determined from the potentiometer output by computing the mean and the within-subject standard deviation in the absolute constant errors (ICEs) for each participant over each set of 10 trials. The constant error (CE) in time to reversal was the measure of temporal accuracy. In order to determine error detection capability, the mean absolute difference between the objective and subjective scores (mean *IO-SI*) was computed for each participant over each set of 10 trials. For acquisition, all dependent variables were analyzed in separate 2 x 9 (Random/Blocked Acquisition Condition x Trial Block) analyses of variance (ANOVAs) with repeated measures on the last factor. For both retention tests, all dependent variables were analyzed with 2 x 2 x 3 (Random/Blocked Acquisition Condition x Random/Blocked Retention Condition x Trial Block) ANOVAs with repeated measures on the last factor. Due to the number of ANOVAs run, and to help control for the experimentwise error rate, the alpha level was set at .01. Also, in order to counter possible violations of the sphericity assumption, all degrees of freedom for *F* ratios involving repeated measures were adjusted using the Huynh-Feldt epsilon coefficient.

Results

Temporal Performance

The mean CE in time to reversal during acquisition for the random practice group and the blocked practice group was 9 ($SD = 23$) and 7 ms ($SD = 23$), respectively. The effect of acquisition condition was not significant. At immediate retention, the mean CE in time to reversal for the groups with random and blocked practice during acquisition was 1 ($SD = 22$) and 3 ms ($SD = 28$), respectively. At delayed retention, the mean CE in time to reversal for the groups with random and blocked practice during acquisition was 3 ($SD = 19$) and 6 ms ($SD = 27$), respectively. The effect of all factors was not

significant for either retention test.

Spatial Performance

Figure 1 shows the mean ICEI for each group for acquisition, immediate retention, and delayed retention. The random practice group displayed greater ICEI than the blocked practice group for acquisition. The effect of acquisition condition was significant, $F(1, 22) = 10.3$, $p < .01$, $\omega^2 = .09$, but the Acquisition Condition $\times$ Trial Block interaction was not significant, $F(5, 114) = 2.2$, $p > .01$. The effect of trial block was also significant, $F(5, 114) = 8.8$, $p < .01$, $\omega^2 = .61$.

At immediate and delayed retention, the blocked practice group showed greater ICEI than the random practice group. The effect of acquisition condition was significant for immediate retention, $F(1, 20) = 11.0$, $p < .01$, $\omega^2 = .22$, but not for delayed retention, $F(1, 20) = 5.0$, $p > .01$. No other factors were significant for either retention test.

During acquisition, the random practice group ($SD = 3.8^\circ$) tended to be more consistent than the blocked practice group ($SD = 4.4^\circ$). However, the effect of acquisition condition was not significant, $F(1, 22) = 5.7$, $p > .01$. At immediate retention, participants engaging

in random practice during acquisition tended to be more consistent ($SD = 3.8^\circ$) than blocked practice participants ($SD = 4.7^\circ$), but the effect of acquisition condition was not significant, $F(1, 20) = 7.5$, $p > .01$. At delayed retention, the random practice group ($SD = 4.4^\circ$) was more consistent than the blocked practice group ($SD = 5.8^\circ$). The effect of acquisition condition was significant, $F(1, 20) = 10.1$, $p < .01$, $\omega^2 = .26$.

Error Detection Capability

The mean IO-SI scores for acquisition, immediate retention, and delayed retention are shown in Figure 2. The mean IO-SI differences were greater in the random practice group during acquisition compared to the blocked practice group. The effect of acquisition condition was significant, $F(1, 22) = 53.5$, $p < .01$, $\omega^2 = .31$. The Acquisition Condition $\times$ Trial Block interaction was not significant, $F(7, 151) = 3.0$, $p > .01$. The effect of trial block was also significant, $F(7, 151) = 10.6$, $p < .01$, $\omega^2 = .45$.

At immediate and delayed retention, the blocked practice group showed greater mean IO-SI scores than the random practice group. The effect of acquisition condition was significant for immediate retention, $F(1, 20) = 12.7$, $p < .01$, $\omega^2 = .25$, and delayed retention, $F(1,$

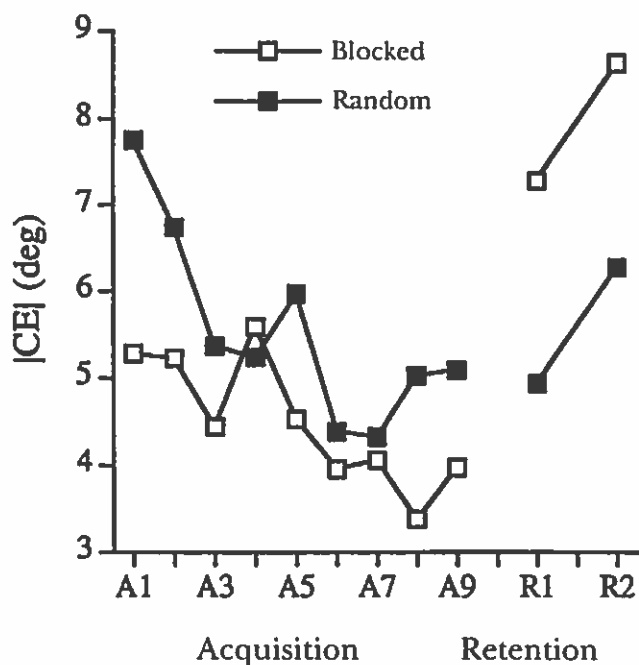


Figure 1. The mean spatial absolute constant error (ICEI) averaged over all amplitudes for acquisition and retention for the random and blocked practice groups in Experiment 1. A1 through A9 are 10-trial acquisition blocks with knowledge of results (KR); R1 is a 10-trial, immediate, no-KR retention block; and R2 is a 10-trial, delayed, no-KR retention block. Random and blocked in the legend refer to acquisition condition. Because the trials were analyzed chronologically, the increase in error at A4 and A7 in the blocked group is probably due to the negative effect of beginning practice on a new amplitude.

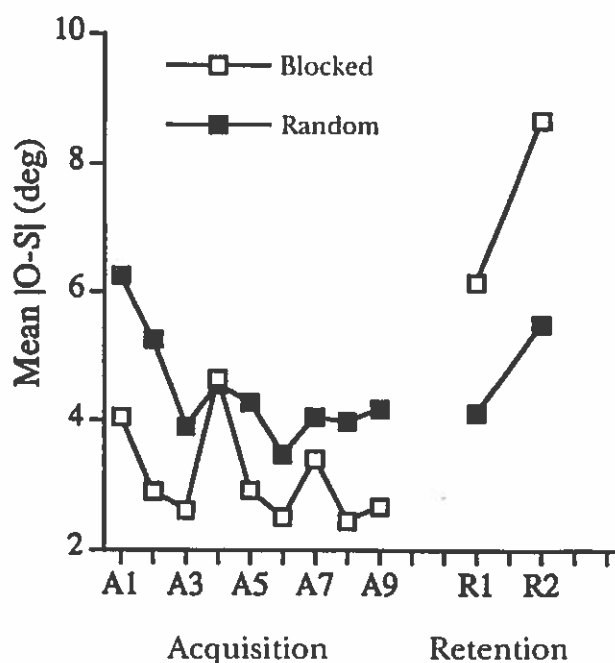


Figure 2. The mean absolute objective-subjective difference (IO-SI) averaged over all amplitudes for acquisition and retention for the random and blocked practice groups in Experiment 1. A1 through A9 are 10-trial acquisition blocks with knowledge of results (KR); R1 is a 10-trial, immediate, no-KR retention block; and R2 is a 10-trial, delayed, no-KR retention block. Random and blocked in the legend refer to acquisition condition. Because the trials were analyzed chronologically, the increase in the mean IO-SI score at A4 and A7 in the blocked group is probably due to the negative effect of beginning practice on a new amplitude.

20) = 11.2, $p < .01$, $\omega^2 = .20$. No other factors were significant for either retention test.

Discussion

The goal of the first experiment was to determine which type of variable practice, random or blocked, was most beneficial for learning a simple motor task and developing a task-specific spatial error detection capability. Blocked practice resulted in better movement accuracy during acquisition compared to random practice. For example, blocked practice participants had lower ICEI compared to random practice participants. However, participants having random practice during acquisition performed better during no-KR retention compared to blocked practice participants in both retention conditions (random or blocked). The findings suggest that random practice, as opposed to blocked practice, is most beneficial for learning a simple spatial-temporal motor task.

The results here also suggest that the benefits of random practice can occur when only the parameter of the motor program is changed from trial to trial rather than the whole program. This finding runs counter to some earlier studies on contextual interference (e.g., Gabriele, Hall, & Buckolz, 1987; Lee & Magill, 1983; Lee, Wulf, & Schmidt, 1992; Pigott & Shapiro, 1984; Wood & Ging, 1991; Wulf & Lee, 1993), so it might be fruitful to explore possible explanations for the current findings. One possible explanation of why the positive benefits of random practice were demonstrated in the present experiment and not in earlier studies could be due to the demands of the practice situation. It has been suggested that random practice with the same motor program has failed to generate retention benefits because the effort required on the part of the participant was low, resulting in a lower level of contextual interference compared to when trial-to-trial variations in the motor program were required (Magill & Hall, 1990). In the present experiment, asking participants to produce a spatially accurate response, in the correct time, and to evaluate their spatial accuracy on each trial could have increased task difficulty and contextual interference to a point where the benefits of random practice would be achieved. It could be that random practice with the same motor program can be more effective than blocked practice as long as practice involves other activities or conditions that increase the overall task difficulty. In the words of Lee, Swinnen, and Serrien (1994), the addition of the error detection task to the random practice manipulation in the present experiment increased the required "cognitive effort" (p. 329) on the part of the participants, resulting in better retention.

The advantage for random variable practice over blocked variable practice in the present experiment could

have also been due to the lack of usefulness of KR during random practice. According to Wulf and Schmidt (1994), participants may pay more attention to intrinsic feedback and less to KR during random practice compared to blocked practice. During acquisition, random practice participants cannot use KR from the previous trial to correct errors on the next trial because a different task or parameter is required. Therefore, KR is not as useful as in blocked practice where KR can help correct errors on the same task on the following trial. In support of this point, Wulf and Schmidt gave reminder KR about the last trial of the same task version during random practice and showed no advantages over a blocked practice group with reminder KR. With KR being less important in random practice conditions, participants may pay more attention to intrinsic feedback which may result in a better long-term retention relative to blocked practice conditions. Incidentally, the fact that random practice participants were more consistent during acquisition than blocked practice participants is congruous with Wulf and Schmidt's notion. Blocked practice participants probably used KR on a previous trial to correct errors on the next trial, resulting in relatively more trial-to-trial variability than in the random practice participants who could not use KR in this way.

The finding that random variable practice is better than blocked variable practice for developing a spatial error detection mechanism suggests that recognition capabilities can also be enhanced with random practice. In fact, the smaller objective-subjective differences in the random participants during retention suggest that the best way to develop recognition capabilities is to use random variable practice during acquisition rather than blocked variable practice.

The major theoretical implication here is that recognition schema strength is enhanced with random variable practice relative to blocked variable practice. Although schema theory (Schmidt, 1975) has difficulty in explaining such a finding, the advantage for random variable practice over blocked variable practice in developing a spatial error detection capability could have been due to the lack of usefulness of KR during random practice (Wulf & Schmidt, 1994). If random practice participants paid more attention to intrinsic feedback and less to KR during acquisition compared to blocked practice participants, then one would expect random practice participants to have better error detection capability compared to blocked practice participants. This was certainly the case at retention when the random practice participants displayed better error detection capability than blocked practice participants. Finally, because random practice participants had a better error detection capability on retention, they probably used that capability to reduce actual movement errors on retention relative to blocked practice participants.

The findings on error detection also extend the cur-

rent literature which had focused on temporal error detection (e.g., Schmidt & White, 1972) to spatial error detection tasks. Although this finding should come as no surprise to teachers, coaches, and performers of motor skills, it is comforting to know that spatial error detection capabilities improve with practice similar to temporal error detection capabilities.

The first experiment was designed as a straightforward contextual interference experiment in order to show the effects of random and blocked practice on spatial performance and error detection capability. The experiment suggests that the increased information-processing demands created by random practice relative to blocked practice apply to both error recognition and movement production processes. One of the major implications is that in order for random variable practice to have a benefit, one must increase the task difficulty (and the contextual interference) by having the participants, for example, guess errors. The second experiment was designed to test this idea by not having participants guess their errors until the no-KR retention test. If random variable practice is beneficial relative to blocked variable practice and participants are not required to perform any additional tasks during acquisition, then there must be a different explanation for the benefits of random variable practice.

Experiment 2

Method

All methods were identical to the first experiment except for the following details.

Participants

The participants were 40 right-handed undergraduate kinesiology students (32 women and 8 men, mean age = 22.6 years, $SD = 6.3$) at the University of Colorado.

Experimental Design

A mixed experimental design was employed with the participants quasi-randomly assigned (to equate for gender) to either random ($n = 20$) or blocked ($n = 20$) practice during acquisition. Participants in the blocked practice group consecutively practiced 24 trials of the three amplitudes (i.e., 20, 40, and 60°), resulting in 72 total practice trials. The order of the amplitudes in the blocked condition was randomly assigned for each participant. Participants in the random practice group practiced a different amplitude on each trial with the restriction that a given amplitude was never repeated on consecutive trials. The 72 trials were analyzed in

blocks of 8 trials, forming the repeated measure factor.

Two additional independent variables were employed at retention. Participants were randomly assigned to either a random or blocked retention practice condition and assigned to either an estimate or no-estimate evaluation condition. Those participants in the estimate condition were asked to estimate their spatial errors on the no-KR retention test. Thus, eight groups with 5 participants in each group (blocked/blocked/estimate, blocked/random/estimate, random/random/estimate, random/blocked/estimate, blocked/blocked/no estimate, blocked/random/no estimate, random/random/no estimate, and random/blocked/no estimate) were formed for the retention test. In addition, a 30° amplitude was added during the retention test as a novel transfer amplitude. Participants performed 40 trials of practice (10 for each amplitude) 2 min following acquisition in an immediate retention test. The 40 trials were analyzed in blocks of 10 trials, forming the repeated measure factor for retention.

Procedure

The major change in the procedure was that participants did not subjectively evaluate their responses during acquisition. Only those participants who were randomly assigned to a subjective evaluation condition at retention were asked to estimate their spatial errors. As in the first experiment, participants were not told that a no-KR retention test was required. Due to the lack of significance for ICEI on the delayed retention test in the first experiment, only an immediate retention test was given in the second experiment.

Data Analysis

For acquisition, all dependent variables were computed for each block of eight trials and analyzed in separate 2 x 9 (Random/Blocked Acquisition Condition x Trial Block) ANOVAs with repeated measures on the last factor. For retention, spatial ICEI and CE in time to reversal were analyzed with 2 x 2 x 2 x 4 (Random/Blocked Acquisition Condition x Random/Blocked Retention Condition x Estimate/No-Estimate Evaluation Condition x Trial Block) ANOVAs with repeated measures on the last factor. The mean IO-SI scores were analyzed with a 2 x 2 x 4 (Random/Blocked Acquisition Condition x Random/Blocked Retention Condition x Trial Block) ANOVA with repeated measures on the last factor.

Results

Temporal Performance

The mean CE in time to reversal during acquisition

was 13.6 ms ($SD = 33$). No effects were significant. At retention, the mean CE in time to reversal was 24.6 ms ($SD = 35$). No effects were significant.

Spatial Performance

Figure 3 shows the mean ICEI for each group and trial block for acquisition and retention. The random practice group displayed greater spatial ICEI than the blocked practice group, especially for the first three trial blocks. The Acquisition Condition $\times$ Trial Block interaction was significant, $F(7, 270) = 4.2$, $p < .01$, $\omega^2 = .11$. An analysis of simple main effects (Kirk, 1968) indicated greater ICEI for the random practice group compared to the blocked practice group on Trial Blocks 1, 2, 3, 5, and 6. The effect of acquisition condition was significant, $F(1, 38) = 28.0$, $p < .01$, $\omega^2 = .12$. The effect of trial block was also significant, $F(7, 270) = 17.6$, $p < .01$, $\omega^2 = .59$. At retention, participants having random practice during acquisition had lower spatial ICEI scores compared to blocked practice participants. The effect of acquisition condition was significant, $F(1, 32) = 7.6$, $p < .01$, $\omega^2 = .12$.

During acquisition, the random practice group ($SD = 3.9^\circ$) was more consistent than the blocked practice group ($SD = 4.5^\circ$). The effect of acquisition condition was significant, $F(1, 38) = 8.0$, $p < .01$, $\omega^2 = .03$. The effect of trial block was also significant, $F(8, 304) = 19.0$, $p < .01$, $\omega^2 = .70$. No effects were significant for retention.

Error Detection Capability

At retention, the mean IO-SI scores for the group with random practice during acquisition ($M = 5.6^\circ$, $SD = 2.2$) were lower than the group with blocked practice during acquisition ($M = 7.6^\circ$, $SD = 3.2$). The effect of acquisition condition was significant, $F(1, 16) = 19.0$, $p < .01$, $\omega^2 = .37$. No other effects were significant.

General Discussion

As in the first experiment, participants with blocked practice performed better than those with random practice during acquisition. However, during retention, a contextual interference effect was again noted with those participants having random practice during acquisition performing with smaller spatial errors compared to blocked practice participants. The results here replicate the finding of the first experiment that contextual interference effects can occur when only the parameter of the motor program is varied.

It is also interesting to note that the benefits of random practice in movement production were found in this experiment when the participants were not re-

quired to evaluate their errors during acquisition. It was suspected, based on results from the first experiment, that the benefits of random parameter practice, instead of practice involving different motor programs, would only be found at retention if task difficulty and contextual interference had somehow been increased during acquisition. This was not the case. It could be that the current task, with spatial and temporal goals, was difficult enough to allow for contextual interference effects to be seen. Perhaps random parameter practice induced the creation of more elaborate or distinctive movement representations that allowed for better spatial accuracy during retention compared to blocked practice (Shea & Zimny, 1983). However, the fact remains that few studies, in addition to this one (e.g., Wulf, 1992), have shown the benefits of random practice over blocked practice with only a parameter—not program variation. It could be that the type of parameter (e.g., overall timing or force) varied in random practice has an influence in the amount of contextual interference and in the strength of the resulting memory representation present during acquisition. Clearly, more work is needed on this issue before any conclusions can be drawn.

Moreover, the benefits of random practice in developing a spatial error detection capability were replicated in the second experiment. Random practice par-

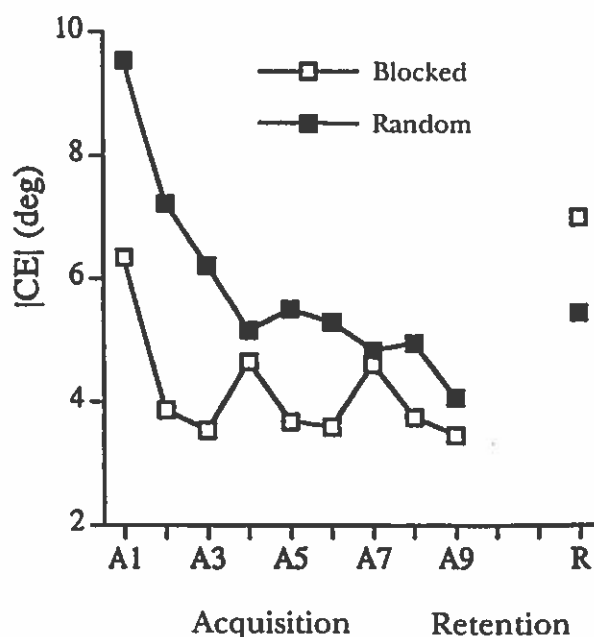


Figure 3. The mean spatial absolute constant error (ICEI) averaged over all amplitudes for acquisition and retention for the random and blocked practice groups in Experiment 2. A1 through A9 are 10-trial acquisition blocks with knowledge of results (KR), and R is a 10-trial, immediate, no-KR retention block. Because the trials were analyzed chronologically, the increase in error at A4 and A7 in the blocked group is probably due to the negative effect of beginning practice on a new amplitude.

ticipants had lower objective-subjective differences than blocked practice participants, providing further evidence that random variable practice is better than blocked variable practice for learning and transfer of recognition capabilities. It is also interesting to note that random practice resulted in better error detection on retention whether or not participants were required to evaluate their errors during acquisition. It could be that random practice facilitates error detection and correction processes relative to blocked practice even when errors are not verbalized. Participants may pay more attention to intrinsic feedback during random practice compared to blocked practice, which would result in a better error detection mechanism. This emphasis on intrinsic feedback may be even more crucial during random practice, where extrinsic feedback (i.e., KR) is not relevant on the next trial as it is for blocked practice (Wulf & Schmidt, 1994). Of course, Shea and Zimny's (1983) elaboration hypothesis, as noted earlier, could also explain the advantages of random practice in the development of a spatial error detection capability.

A final point to be addressed is the consequence of using a more rigorous alpha level (i.e., .01) than usual. With a more stringent alpha level, several interactions were not significant that might have been significant under more lenient circumstances. One interesting, but nonsignificant ($.05 > p > .01$), interaction occurred between acquisition condition and guess for spatial ICEI at retention in Experiment 2. The finding was that participants using random practice during acquisition had a lower spatial ICEI at retention than participants with blocked acquisition practice only when they were required to guess spatial errors. For those participants who were not required to guess spatial errors on retention in Experiment 2, no advantage for random practice was found. The addition of the guessing task during retention could be viewed as a secondary task which, when combined with the movement task, increased errors on the main task for the blocked practice participants, but not for the random practice participants. Perhaps the blocked practice participants were unprepared to evaluate response-produced feedback and to estimate errors during retention because they did not engage in the same level of covert error detection or pay as much attention to intrinsic feedback during the acquisition phase compared to the random participants. If this Acquisition Condition $\times$ Guess interaction is replicated in later studies, more credence can be given to the thoughts here.

In summary, across both experiments, the fact that random practice benefited both movement performance and error detection capability suggests that the increased information-processing demands created by random practice generalize to both error recognition and movement production processes.

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Author's Notes

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